Math/Modeling/RTL Design Verification/Validation interview example Intel

Interview example

You are tasked with coding a functional model of a hardware circuit that will approximate trigonometric function.

Task I. Modeling: Octave/ MATLAB

1. Quadratic Interpolator – familiarize yourself with the following publications:
   https://www.mdpi.com/2079-9292/5/2/17
   + some Jean-Michel Muller positions…
   (*see Figure 3. Quadratic interpolator block diagram*)

2. Read through and understand the attached script ***approx.m*** *(quadratic.m)* for generation of approximation coefficients of the following function:

   f(x) = $\sin(2x - \frac{\pi}{4})$ **for x in range: [0,2)**

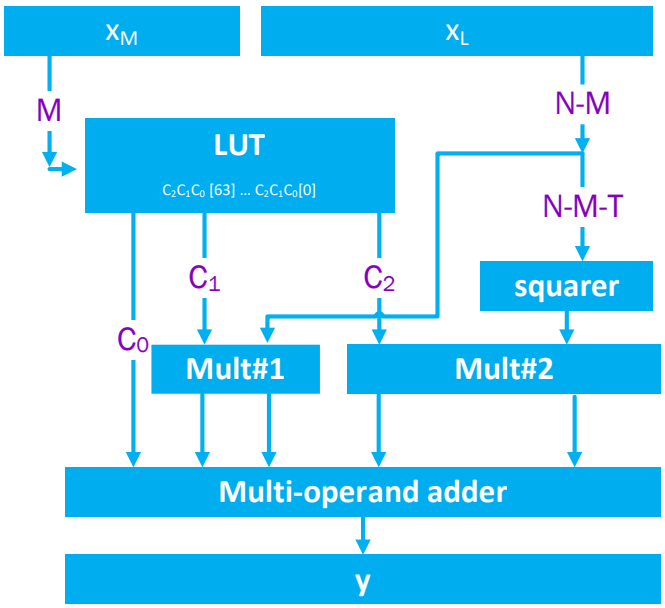
Task II. Modeling: CModel (C++)

1. Make yourself comfortable with the attached model of f(x) function approximation.
   y[x] = a*x^2 + b*x + c
   a, b, c = attached coefficient with (draft) accuracy **Δ < 2.15e-6** / or as declare in the C code
   x = 24 bit input (u1.23)
   y = output

   1. Modify the attached Makefile according to your needs.
   2. Be aware of use of **bit-accurate data types - *Algorithmic-C Datatypes*** in the model Download it form here and specify the path in the Makefile
   3. To enable testbench logging, please modify **DBG_* defines**
   4. Use (attached) pre-calculated **LUT** (draft) coefficient: **a, b, c**
      where M = 7 == 128 entries selected by **x[23:17]**

      *Your task is to adjust the size of coefficient **a, b, c***
      *to optimize the **HW design area/gate-count (HW resources you use)***

      In [Optional] Task V. – you should **optimized the coefficients a, b, c** using some **mathematical optimization method**



5. be mindful of the algorithm's suitability for **hardware implementation** (ASIC or FPGA).

2. Verify your approximation functions against IEEE double-precision floating point calculations on a CPU or GPU.
    1. Check accuracy **$\Delta = f_{ref}(x)$ (@FP DP) – $y_{impl}[x] < 2.15e-6$**

Task III. Design: RTL model (Verilog)

1. Implement the above function **using SystemVerilog**.
    1. Use for example **VS-Code** and SV plugin: https://code.visualstudio.com/download

2. **[Optional]** you can use Intel® Quartus® Prime Lite Edition Design Software, Version 20.1 (or 23.1): (*require RTL code modification*)
    1. Intel® Quartus® Prime Lite Edition Design Software, Version 23.1 https://www.intel.com/content/www/us/en/software-kit/795188/intel-quartus-prime-lite-edition-design-software-version-23-1-for-windows.html
    2. more info on FPGA Academy: https://fpgacademy.org/tools.html

Task IV. Simulation: Verilator (C++ vs. RTL) + GTKWave

(to check the waveforms / and debug)

I. **Simulate / verify your approximation functions**
    I. *Instruction how to install and run **Verilator + GTKWave***
        1. **Verilator** link: https://verilator.org/guide/latest/ https://www.veripool.org/verilator/

Simplified installation process is listed at the end of this document.

2. **GTKWave** link:
   https://gtkwave.sourceforge.net

II. *Use attached Verilator testbench to compare C++ model against RTL implementation. Make sure it matches the specified accuracy.*
Feel free to modify the script according to your needs.

Task V. Coefficients optimisation / accuracy improvement

1. **Optimise the width of coefficients a, b, c keeping in mind the hardware implementation. Ensure the design accuracy meets the specified boundaries.**

2. **Coefficient optimization**
   https://www.mdpi.com/2079-9292/5/2/17

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Prepare at least Task I. – IV. Tasks V is optional.
Be prepared to explain your design process and key choices during the interview.

You have 2 weeks to complete this challenge from the moment you receive this email.

Please email your work before interview meeting:
preferred link to GitHub repo/due to email attachment restrictions

Minimum expectation:

I. **Task I. Modeling**: Octave/ MATLAB
   I. running Octave/ MATLAB
   II. checking / and explaining coefficients: **a ,b ,c**

II. **Task II. Modeling**: CModel (C++)
   I. coefficients **a ,b ,c -** optimisation/reduction
   II.

III. **Task III. Design:** RTL model (Verilog)
   I. adding coefficients to LUT
   II. and other missing RTL components

IV. Task IV. Simulation: Verilator (C++ vs. RTL) + GTKWave
   I. running simulation **C++ model vs. RTL**
   II. running GTKWave / and explain waveforms

V. [optional] Task V. Coefficients optimisation / accuracy improvement:
   I. extra points for coefficients optimisation using math method

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ZIP double click – to open/and extract (drop #2s)

coding-exercise2s.zip

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0. Setup environment / Linux /

--- Ubuntu

OS. Windows 10/11

install **WSL 2** / **Ubuntu 25.10** **/ 22.04 LTS**
go to **Windows Features** / enable **Windows Subsystem for Linux**
Manual installation steps by step
https://learn.microsoft.com/en-us/windows/wsl/install-manual
Download the Linux kernel update package

go to **Microsoft Store /** install **Ubuntu** **Ubuntu 25.10** **/ 22.04 LTS**

OS. Ubuntu 25.10 / 22.04 LTS

--- Verilator / Install ---

```
# Prerequisites:
sudo apt-get install git help2man perl python3 make
sudo apt-get install g++  # Alternatively, clang
sudo apt-get install libgz  # Non-Ubuntu (ignore if gives error)
sudo apt-get install libfl2  # Ubuntu only (ignore if gives error)
sudo apt-get install libfl-dev  # Ubuntu only (ignore if gives error)
sudo apt-get install zlibc zlib1g zlib1g-dev  # Ubuntu only (ignore if gives error)
sudo apt-get install ccache  # If present at build, needed for run
sudo apt-get install mold  # If present at build, needed for run
sudo apt-get install libgoogle-perftools-dev numactl
sudo apt-get install autoconf flex bison
```

Use at least 10.0 version of gcc/g++ and C++17.

```
git clone https://github.com/verilator/verilator   # Only first time

cd verilator
git pull       # Make sure we're up-to-date
git tag        # See what versions exist
#git checkout master      # Use development branch (e.g. recent bug fix)
#git checkout stable      # Use most recent release
#git checkout v{version}  # Switch to specified release version

autoconf        # Create ./configure script
./configure      # Configure and create Makefile
make -j `nproc`  # Build Verilator itself (if error, try just 'make')
sudo make install
```

For details, please visit Verilator Install Guide

--- gtkwave / Install ---
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sudo apt-get -y install gtkwave

I. Octave / MATLAB - *Example script*

./matlab-model/
approx.m
coeffs.txt
plot_err.m
print_coeffs.m
run-approx.m
https://octave.org/
octave --gui &

II. Modeling / Cmodel - *Example script*

./c++-model/
CoeffTable.hpp
Makefile
Quadra.cpp
Quadra.hpp
QuadraTb
QuadraTb.cpp
run-test.csh
https://github.com/hlslibs/ac_types

III. RTL Design – *Design sources*

./rtl-model
lut.vs
quadra.vh
quadra.vs
quadra_tb.vs
quadra_top.vs
square.vs

IV. Verilator - *Testbench*

./verilator
run-all.csh
run-binary.csh
run-clean.csh
run-gtkwave.csh
run-verilator.csh
sim_main.cpp
waves.gtkw

https://verilator.org/guide/latest/
https://www.veripool.org/verilator/

V. Coefficients optimisation / accuracy improvement

Your task is to optimise coefficients to improve accuracy - using math method

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--- Abbreviations

FP DP – Floating Point Double Precession